



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO CPS AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

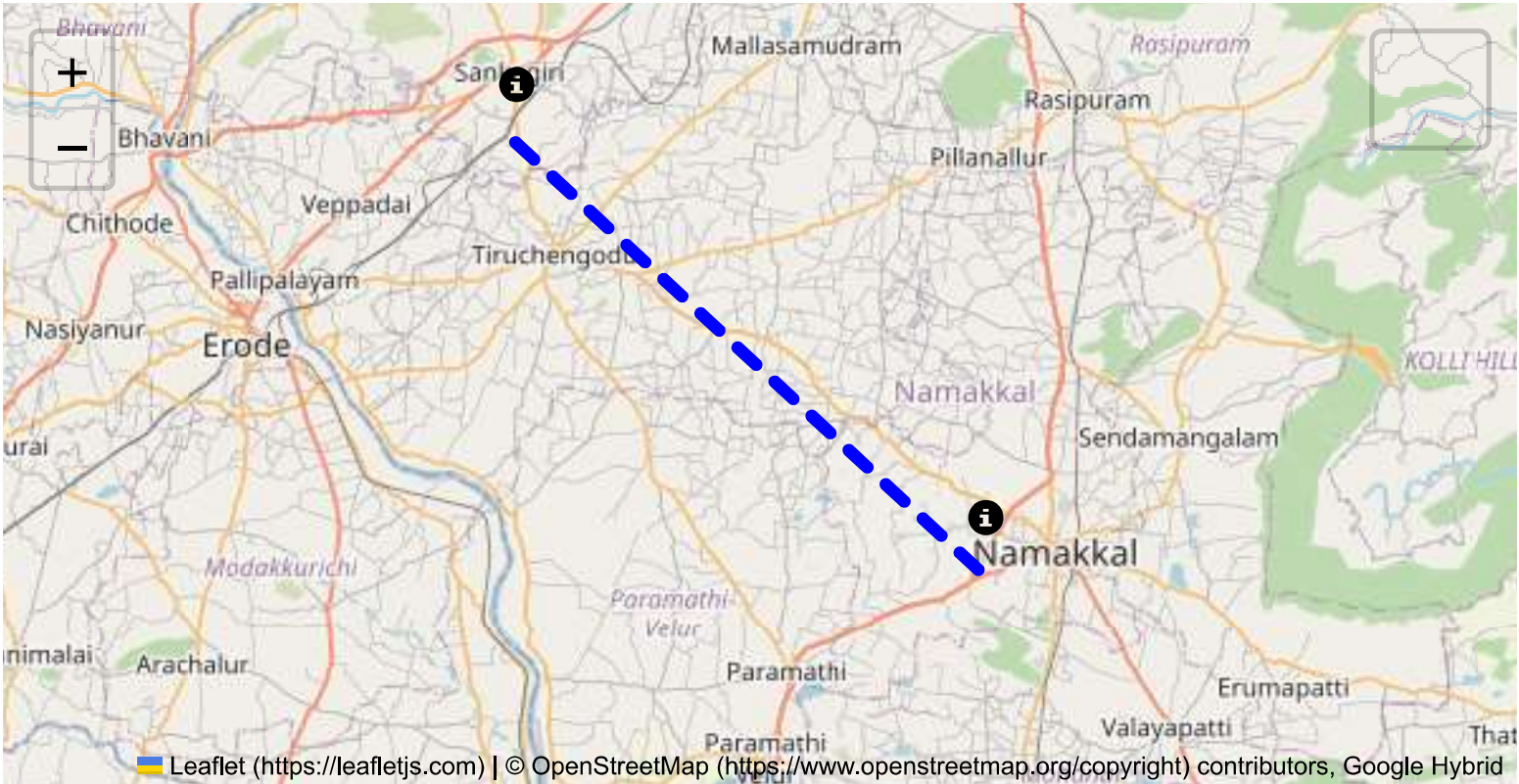
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 47.41 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.207158, 78.129882)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Vallipuram Post in Paramathi Rd, Tamil Nadu spans approximately 47.41 kilometers. The journey typically takes around 1.03 hours for heavy vehicles carrying hazardous materials. The route might include urban as well as rural road segments, serving as the primary connection between these locations.

Typical Weather Conditions and Hazards

Tamil Nadu typically experiences a tropical climate. The key weather-related hazards during this route include:

- **Monsoon Season (June to September):** Heavy rainfall could lead to slippery roads and reduced visibility.
- **Summer (March to May):** High temperatures can cause asphalt roads to become soft, potentially affecting vehicle stability.
- **Cyclones:** Although infrequent, cyclones during October to December could severely impact travel safety.

Traffic Patterns

- **Peak Hours:** Morning (8:00 AM - 10:00 AM) and Evening (5:00 PM - 7:00 PM), particularly near town centers and junctions.
- **Congestion-Prone Areas:** Key intersections and near major towns where local traffic intermingles. Watch out for market areas and bus stops where pedestrian flow can slow down traffic.

Road Quality and Infrastructure

- **Urban Sections:** Generally well-maintained but may suffer from congestion and narrow lanes.
- **Rural Sections:** Potential for potholes and uneven surfaces. Be cautious of unmarked speed bumps and animal crossings.

Alternative Routes for Emergencies

In case of route blockages or emergencies, consider using NH44 and connecting regional highways that may run parallel to the primary route. Local routes can also offer diversions, though they may be less direct and not suitable for heavy vehicles.

Local Regulations on Hazardous Material Transport

Ensure compliance with:

- **Tamil Nadu Traffic Police Regulations:** Check for any local permits needed.
- **Transport Department Notifications:** Regular updates on transportation of hazardous goods might necessitate specific actions or route changes.

Historical Incidents

While specific incidents are not detailed here, it is wise to check local news archives for reports that mention any road accidents involving freight vehicles, especially those carrying hazardous materials.

Environmental Considerations

Be mindful of:

- **Wildlife Crossings:** Certain stretches may pass through areas with wildlife activity.
- **Water Bodies and Agricultural Land:** Avoid contaminating wetlands or farmlands in the event of a spill.

Communication Coverage

Urban areas along the route generally have good mobile network coverage. However, rural segments might face weak signals or dead zones, particularly in less developed sections.

Estimated Emergency Response Times

- **Urban Areas:** Response times are typically faster, averaging around 15-20 minutes.
- **Rural Areas:** Response might be delayed due to distance from emergency services, potentially taking 30 minutes or more.

Overall Summary of Risk Assessment

This route does pose moderate risk, factoring in weather conditions, road quality, and traffic patterns. Drivers must be cautious during monsoon seasons and remain aware of potential traffic congestion. Proper planning and compliance with local regulations can mitigate many risks, ensuring safe passage for heavy vehicles carrying hazardous material. Regular updates on road conditions and proactive communication with local transport authorities will enhance route safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	11.20387, 78.12203	15 KM/Hr
2	U-Turn	High	11.2089758, 78.1355753	10 KM/Hr
3	Turn	High	11.43811, 77.87348	15 KM/Hr
4	Turn	Medium	11.43956, 77.87340	30 KM/Hr
5	Turn	High	11.43968, 77.87345	15 KM/Hr
6	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
7	Turn	Medium	11.38342, 77.89975	30 KM/Hr
8	Turn	Medium	11.38190, 77.90254	30 KM/Hr
9	Turn	High	11.38186, 77.90258	15 KM/Hr
10	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
11	Blind Spot	Blind Spot	11.23936, 78.14267	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.22817, 78.13576	15 KM/Hr
13	Turn	High	11.22822, 78.13561	15 KM/Hr
14	Turn	Medium	11.20882, 78.12464	30 KM/Hr
15	Blind Spot	Blind Spot	11.20898, 78.13558	10 KM/Hr
16	Turn	High	11.20890, 78.13558	15 KM/Hr
17	Turn	High	11.20761, 78.13016	15 KM/Hr
18	Turn	High	11.20730, 78.13013	15 KM/Hr
19	Turn	High	11.20721, 78.13001	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.20387, 78.12203



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.2089758, 78.1355753



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.23936, 78.14267



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.22817, 78.13576



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.22822, 78.13561



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.20882, 78.12464



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.20898, 78.13558



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.20890, 78.13558



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.20761, 78.13016



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.20730, 78.13013



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.20721, 78.13001